

Batch: DMW 2 Roll No.: 16010324044
Experiment/assignment/tutorial No.: 08
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Implementation of PostgreSQL commands

AIM: To learn PostgreSQL commands required for database creation and manipulation

OUTCOME: Students will be able to build a data warehouse for OLAP

Procedure:

Refer to websites: <https://neon.com/postgresql/tutorial>

<https://www.w3schools.com/postgresql/>

& PostgreSQL cheatsheets provided to learn various Data Definition Language(DDL) commands, Data Manipulation Language(DML) commands, Data Query Language(DQL) commands, Data Control Language(DCL) commands, Transaction Control Language(TCL) commands

1. Create a table emp with following schema:

```
(empno , empname,DOB ,salary ,designation )
```

```
Add values: 100,John,4.21.1994, 50000,Manager  
101,Greg,6.20.1994,25000,Clerk
```

4. Display all values of the table
5. Display only empno and empname from table

6. Create table manager:

```
(enumber ,ename ,salary ,commission ,annualsalary ,hiredate,designation ,deptno ,reporting )
```

7. Add values : 7369,Dharsan,2500,500,30000,30-June-81,clerk,10,John
7839,Subu,3000,400,36000,1-Jul-82,manager,null,James
7934,Aadhi,3500,300,42000,1-May-82,manager,30,NULL
7788,Vikash,4000,0,48000,12-Aug-82,clerk,50,Bond

8. Update all the records of manager table by increasing 10% of their salary as bonus.
9. Delete the records from manager table where the salary less than 2750
10. Display each name of the employee as "Name" and annual salary as "Annual Salary" (Note: Salary in emp table is the monthly salary)
11. List concatenated value of name and designation of each employee

12. List names of employees whose employee numbers are 7369,7839,7934,7788
13. List the names of employee who are not Managers.
14. List the names of employees not belonging to dept no 30,40 & 10
15. List names of those employees joined between 30 June 81 and 31 Dec 81
16. List different designations in the company.

17. List names and designations of employee who does not report to anybody
18. List employees whose name either start or end with 's'
19. Sort emp table in ascending order by hire-date and list ename, job, deptno and hire-date
20. Sort emp table in descending order by annual salary and list empno, ename, job and annual-salary

21. Create Table ACTOR :

(ACT_ID (PRIMARY KEY),ACT_NAM, ACT_GENDER)

22. Create Table DIRECTOR:

(DIR_ID (PRIMARY KEY),DIR_NAME ,DIR_PHONE)

23. Create table MOVIES:

**(MOV_ID(PRIMARY KEY),MOV_TITLE ,MOV_YEAR ,MOV_LANG ,
DIR_ID ,FOREIGN KEY (DIR_ID) REFERENCES DIRECTOR(DIR_ID))**

24. Create table MOVIE_CAST:

**(ACT_ID,MOV_ID,ROLE,PRIMARY KEY (ACT_ID,MOV_ID),FOREIGN KEY (ACT_ID) REFERENCES
ACTOR(ACT_ID),FOREIGN KEY (MOV_ID) REFERENCES MOVIES(MOV_ID))**

25. Create table RATING:

**(MOV_ID (PRIMARY KEY), REV_STARS, FOREIGN KEY (MOV_ID) REFERENCES
MOVIES(MOV_ID))**

27. Enter following data into ACTOR table:

```
(101, 'RAHUL', 'M');
(102, 'ANKITHA', 'F');
(103, 'RADHIKA', 'F');
(104, 'CHETHAN', 'M');
(105, 'VIVAN', 'M');
```

28. Enter following data into DIRECTOR table

```
(201, 'ANUP', 918181818);  
(202, 'HITCHCOCK', 918181812);  
(203, 'SHASHANK', 918181813);  
(204, 'STEVEN SPIELBERG', 918181814);  
(205, 'ANAND', 918181815);
```

29. Enter following data into MOVIES table

```
1001, 'MANASU', 2017, 'KANNADA', 201);  
1002, 'AAKASHAM', 2015, 'TELUGU', 202);  
1003, 'KALIYONA', 2008, 'KANNADA', 201);  
1004, 'WAR HORSE', 2011, 'ENGLISH', 204);  
1005, 'HOME', 2012, 'ENGLISH', 205);
```

30. Enter following data into MOVIE_CAST table

```
101, 1002, 'HERO');  
101, 1001, 'HERO');  
103, 1003, 'HEROINE'  
103, 1002, 'GUEST');  
104, 1004, 'HERO');
```

31. Enter following data into RATING table

```
1001, 4)  
1002, 2)  
1003, 5)  
1004, 4)  
1005, 3)
```

32. List the titles of all movies directed by 'Hitchcock'

33. Find the movie names where one or more actors acted in two or more movies.
34. List all actors who acted in a movie before 2000 and also in a movie after 2015 (use JOIN operation)
35. Find the title of movies and number of stars for each movie that has at least one rating --and find the highest number of stars that movie received. Sort the result by --movie title.

Observation number	SQL Query	Output																		
1	<pre>CREATE TABLE emp (empno INT PRIMARY KEY, empname VARCHAR(50), DOB DATE, salary NUMERIC(10,2), designation VARCHAR(30)); INSERT INTO emp (empno, empname, DOB, salary, designation) VALUES (100, 'John', '1994-04-21', 50000, 'Manager'), (101, 'Greg', '1994-06-20', 25000, 'Clerk');</pre>	CREATE TABLE Query returned successfully in 38 msec. <table><tr><th></th><th>empno [PK] integer</th><th>empname character varying (50)</th><th>dob date</th><th>salary numeric (10,2)</th><th>designation character varying (30)</th></tr><tr><td>1</td><td>100</td><td>John</td><td>1994-04-21</td><td>50000.00</td><td>Manager</td></tr><tr><td>2</td><td>101</td><td>Greg</td><td>1994-06-20</td><td>25000.00</td><td>Clerk</td></tr></table>		empno [PK] integer	empname character varying (50)	dob date	salary numeric (10,2)	designation character varying (30)	1	100	John	1994-04-21	50000.00	Manager	2	101	Greg	1994-06-20	25000.00	Clerk
	empno [PK] integer	empname character varying (50)	dob date	salary numeric (10,2)	designation character varying (30)															
1	100	John	1994-04-21	50000.00	Manager															
2	101	Greg	1994-06-20	25000.00	Clerk															
2	SQL <pre>SELECT empno, empname FROM emp;</pre>	<table><tr><th></th><th>empno [PK] integer</th><th>empname character varying (50)</th></tr><tr><td>1</td><td>100</td><td>John</td></tr><tr><td>2</td><td>101</td><td>Greg</td></tr></table>		empno [PK] integer	empname character varying (50)	1	100	John	2	101	Greg									
	empno [PK] integer	empname character varying (50)																		
1	100	John																		
2	101	Greg																		
3	SQL <pre>CREATE TABLE manager (</pre>	CREATE TABLE Query returned successfully in 41 msec.																		

```

enumber INT PRIMARY KEY,
ename VARCHAR(50),
salary NUMERIC(10,2),
commission NUMERIC(10,2),
annualsalary NUMERIC(10,2),
hiredate DATE,
designation VARCHAR(30),
deptno INT,
reporting VARCHAR(50)
);

```

4

SQL

```

INSERT INTO manager

(enumber, ename, salary,
commission, annualsalary,
hiredate, designation, deptno,
reporting)

VALUES

```

enumber	ename	salary	commission	annualsalary	hiredate	designation	deptno	reporting
7369	Oracle	2900.00	10.00	31800.00	1981-05-01	Manager	10	John
7369	Sales	3000.00	400.00	36000.00	1982-01-01	Manager	30	James
7369	Admin	3000.00	300.00	36000.00	1982-05-01	Manager	30	James
7369	Vinay	4000.00	0.00	48000.00	1982-08-01	Clk	50	Bond

	<pre>(7369, 'Dharsan', 2500, 500, 30000, '1981-06-30', 'Clerk', 10, 'John'), (7839, 'Subu', 3000, 400, 36000, '1982-07-01', 'Manager', NULL, 'James'), (7934, 'Aadhi', 3500, 300, 42000, '1982-05-01', 'Manager', 30, NULL), (7788, 'Vikash', 4000, 0, 48000, '1982-08-12', 'Clerk', 50, 'Bond');</pre> <pre>SELECT * FROM manager;</pre>																																														
5	SQL <pre>UPDATE manager SET salary = salary * 1.10; SELECT * FROM manager;</pre>	<table><tr><th>enumber PK integer</th><th>ename character varying (25)</th><th>salary numeric (10,2)</th><th>commission numeric (10,2)</th><th>annualsalary numeric (10,2)</th><th>hiredate date</th><th>designation character varying (25)</th><th>deptno integer</th><th>reporting character varying (25)</th></tr><tr><td>1</td><td>7369 Dharsan</td><td>2750.00</td><td>500.00</td><td>30000.00</td><td>1981-06-30</td><td>Clerk</td><td>10</td><td>John</td></tr><tr><td>2</td><td>7839 Subu</td><td>3300.00</td><td>400.00</td><td>36000.00</td><td>1982-07-01</td><td>Manager</td><td>30</td><td>James</td></tr><tr><td>3</td><td>7934 Aadhi</td><td>3850.00</td><td>300.00</td><td>42000.00</td><td>1982-05-01</td><td>Manager</td><td>30</td><td>Null</td></tr><tr><td>4</td><td>7788 Vikash</td><td>4400.00</td><td>0.00</td><td>48000.00</td><td>1982-08-12</td><td>Clerk</td><td>50</td><td>Bond</td></tr></table>	enumber PK integer	ename character varying (25)	salary numeric (10,2)	commission numeric (10,2)	annualsalary numeric (10,2)	hiredate date	designation character varying (25)	deptno integer	reporting character varying (25)	1	7369 Dharsan	2750.00	500.00	30000.00	1981-06-30	Clerk	10	John	2	7839 Subu	3300.00	400.00	36000.00	1982-07-01	Manager	30	James	3	7934 Aadhi	3850.00	300.00	42000.00	1982-05-01	Manager	30	Null	4	7788 Vikash	4400.00	0.00	48000.00	1982-08-12	Clerk	50	Bond
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	SQL DELETE FROM manager WHERE salary < 2750; SELECT * FROM manager;																
7	SQL SELECT ename AS "Name", annualsalary AS "Annual Salary" FROM manager;	<table><tr><th></th><th>Name character varying (50) 🔒</th><th>Annual Salary numeric (10,2) 🔒</th></tr><tr><td>1</td><td>Dharsan</td><td>30000.00</td></tr><tr><td>2</td><td>Subu</td><td>36000.00</td></tr><tr><td>3</td><td>Aadhi</td><td>42000.00</td></tr><tr><td>4</td><td>Vikash</td><td>48000.00</td></tr></table>		Name character varying (50) 🔒	Annual Salary numeric (10,2) 🔒	1	Dharsan	30000.00	2	Subu	36000.00	3	Aadhi	42000.00	4	Vikash	48000.00
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2	Subu	36000.00															
3	Aadhi	42000.00															
4	Vikash	48000.00															
8	SQL SELECT ename FROM manager WHERE enumber IN (7369, 7839, 7934, 7788);	<table><tr><th></th><th>ename character varying (50) 🔒</th></tr><tr><td>1</td><td>Dharsan</td></tr><tr><td>2</td><td>Subu</td></tr><tr><td>3</td><td>Aadhi</td></tr><tr><td>4</td><td>Vikash</td></tr></table>		ename character varying (50) 🔒	1	Dharsan	2	Subu	3	Aadhi	4	Vikash					
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	ename character varying (50) 🔒																
1	Dharsan																
2	Vikash																

	<pre>WHERE designation <> 'Manager';</pre>							
10	<pre>SQL SELECT ename FROM manager WHERE deptno NOT IN (30, 40, 10);</pre>	<table><tr><td></td><td>ename character varying (50) </td></tr><tr><td>1</td><td>Vikash</td></tr></table>		ename character varying (50)	1	Vikash		
	ename character varying (50)							
1	Vikash							
11	<pre>SQL SELECT ename FROM manager WHERE hiredate BETWEEN '1981-06-30' AND '1981-12-31';</pre>	<table><tr><td></td><td>ename character varying (50) </td></tr><tr><td>1</td><td>Dharsan</td></tr></table>		ename character varying (50)	1	Dharsan		
	ename character varying (50)							
1	Dharsan							
12	<pre>SQL SELECT DISTINCT designation FROM manager;</pre>	<table><tr><td></td><td>designation character varying (30) </td></tr><tr><td>1</td><td>Manager</td></tr><tr><td>2</td><td>Clerk</td></tr></table>		designation character varying (30)	1	Manager	2	Clerk
	designation character varying (30)							
1	Manager							
2	Clerk							
13		<table><tr><td></td><td>ename character varying (50) </td><td>designation character varying (30) </td></tr><tr><td>1</td><td>Aadhi</td><td>Manager</td></tr></table>		ename character varying (50)	designation character varying (30)	1	Aadhi	Manager
	ename character varying (50)	designation character varying (30)						
1	Aadhi	Manager						

	SQL SELECT ename, designation FROM manager WHERE reporting IS NULL;																																																			
14	SQL SELECT ename FROM manager WHERE ename ILIKE 's%' OR ename ILIKE '%s';	<table><tr><td></td><td>ename character varying (50) 🔒</td></tr><tr><td>1</td><td>Subu</td></tr></table>		ename character varying (50) 🔒	1	Subu																																														
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15	SQL SELECT ename, designation, deptno, hiredate FROM manager ORDER BY hiredate ASC; SQL SELECT enumber,	<table><tr><td></td><td>ename character varying (50) 🔒</td><td>designation character varying (30) 🔒</td><td>deptno integer 🔒</td><td>hiredate date 🔒</td></tr><tr><td>1</td><td>Dharsan</td><td>Clerk</td><td>10</td><td>1981-06-...</td></tr><tr><td>2</td><td>Aadhi</td><td>Manager</td><td>30</td><td>1982-05-...</td></tr><tr><td>3</td><td>Subu</td><td>Manager</td><td>[null]</td><td>1982-07-...</td></tr><tr><td>4</td><td>Vikash</td><td>Clerk</td><td>50</td><td>1982-08-...</td></tr></table> <table><tr><td></td><td>enumber [PK] integer 🔗</td><td>ename character varying (50) 🔗</td><td>designation character varying (30) 🔗</td><td>annualsalary numeric (10,2) 🔗</td></tr><tr><td>1</td><td>7788</td><td>Vikash</td><td>Clerk</td><td>48000.00</td></tr><tr><td>2</td><td>7934</td><td>Aadhi</td><td>Manager</td><td>42000.00</td></tr><tr><td>3</td><td>7839</td><td>Subu</td><td>Manager</td><td>36000.00</td></tr><tr><td>4</td><td>7369</td><td>Dharsan</td><td>Clerk</td><td>30000.00</td></tr></table>		ename character varying (50) 🔒	designation character varying (30) 🔒	deptno integer 🔒	hiredate date 🔒	1	Dharsan	Clerk	10	1981-06-...	2	Aadhi	Manager	30	1982-05-...	3	Subu	Manager	[null]	1982-07-...	4	Vikash	Clerk	50	1982-08-...		enumber [PK] integer 🔗	ename character varying (50) 🔗	designation character varying (30) 🔗	annualsalary numeric (10,2) 🔗	1	7788	Vikash	Clerk	48000.00	2	7934	Aadhi	Manager	42000.00	3	7839	Subu	Manager	36000.00	4	7369	Dharsan	Clerk	30000.00
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	<pre> ename, designation, annualsalary FROM manager ORDER BY annualsalary DESC; </pre>	
B*16	<pre> SQL CREATE TABLE actor (act_id INT PRIMARY KEY, act_nam VARCHAR(50), act_gender CHAR(1)); </pre>	<pre> CREATE TABLE Query returned successfully in 38 msec. </pre>
17	<pre> SQL CREATE TABLE director (dir_id INT PRIMARY KEY, dir_name VARCHAR(100), dir_phone BIGINT); </pre>	<pre> CREATE TABLE Query returned successfully in 25 msec. </pre>

18	SQL <pre>CREATE TABLE movies (mov_id INT PRIMARY KEY, mov_title VARCHAR(100), mov_year INT, mov_lang VARCHAR(30), dir_id INT, FOREIGN KEY (dir_id) REFERENCES director(dir_id));</pre>	CREATE TABLE Query returned successfully in 29 msec.
19	SQL <pre>CREATE TABLE movie_cast (act_id INT, mov_id INT, role VARCHAR(50), PRIMARY KEY (act_id, mov_id), FOREIGN KEY (act_id) REFERENCES actor(act_id),</pre>	CREATE TABLE Query returned successfully in 26 msec.

	<pre> FOREIGN KEY (mov_id) REFERENCES movies(mov_id)); </pre>	
20	<pre> SQL CREATE TABLE rating (mov_id INT PRIMARY KEY, rev_stars INT, FOREIGN KEY (mov_id) REFERENCES movies(mov_id)); </pre>	<pre> CREATE TABLE Query returned successfully in 35 msec. </pre>
21	<pre> SQL INSERT INTO actor (act_id, act_nam, act_gender) VALUES (101, 'RAHUL', 'M'), (102, 'ANKITHA', 'F'), (103, 'RADHIKA', 'F'), </pre>	<pre> INSERT 0 5 Query returned successfully in 25 msec. </pre>

	<pre>(104, 'CHETHAN', 'M'), (105, 'VIVAN', 'M');</pre>	
22	SQL <pre>INSERT INTO director (dir_id, dir_name, dir_phone) VALUES (201, 'ANUP', 918181818), (202, 'HITCHCOCK', 918181812), (203, 'SHASHANK', 918181813), (204, 'STEVEN SPIELBERG', 918181814), (205, 'ANAND', 918181815);</pre>	INSERT 0 5 Query returned successfully in 29 msec.
23	SQL <pre>INSERT INTO movies (mov_id, mov_title, mov_year, mov_lang, dir_id) VALUES (1001, 'MANASU', 2017, 'KANNADA', 201),</pre>	INSERT 0 5 Query returned successfully in 25 msec.

	<pre>(1002, 'AAKASHAM', 2015, 'TELUGU', 202), (1003, 'KALIYONA', 2008, 'KANNADA', 201), (1004, 'WAR HORSE', 2011, 'ENGLISH', 204), (1005, 'HOME', 2012, 'ENGLISH', 205);</pre>	
24	<pre>SQL INSERT INTO movie_cast (act_id, mov_id, role) VALUES (101, 1002, 'HERO'), (101, 1001, 'HERO'), (103, 1003, 'HEROINE'), (103, 1002, 'GUEST'), (104, 1004, 'HERO');</pre>	<pre>INSERT 0 5 Query returned successfully in 25 msec.</pre>

25	SQL <pre>INSERT INTO rating (mov_id, rev_stars) VALUES (1001, 4), (1002, 2), (1003, 5), (1004, 4), (1005, 3); SELECT * FROM actor; SELECT * FROM director; SELECT * FROM movies; SELECT * FROM movie_cast; SELECT * FROM rating;</pre>	<table><tr><th></th><th>mov_id [PK] integer</th><th>rev_stars integer</th></tr><tr><td>1</td><td>1001</td><td>4</td></tr><tr><td>2</td><td>1002</td><td>2</td></tr><tr><td>3</td><td>1003</td><td>5</td></tr><tr><td>4</td><td>1004</td><td>4</td></tr><tr><td>5</td><td>1005</td><td>3</td></tr></table>		mov_id [PK] integer	rev_stars integer	1	1001	4	2	1002	2	3	1003	5	4	1004	4	5	1005	3
	mov_id [PK] integer	rev_stars integer																		
1	1001	4																		
2	1002	2																		
3	1003	5																		
4	1004	4																		
5	1005	3																		
26	SQL <pre>SELECT m.mov_title FROM movies m JOIN director d ON m.dir_id = d.dir_id WHERE d.dir_name = 'HITCHCOCK';</pre>	<table><tr><th></th><th>mov_title character varying (100)</th></tr><tr><td>1</td><td>AAKASHAM</td></tr></table>		mov_title character varying (100)	1	AAKASHAM														
	mov_title character varying (100)																			
1	AAKASHAM																			

27	SQL <pre>SELECT DISTINCT m.mov_title FROM movies m JOIN movie_cast mc ON m.mov_id = mc.mov_id WHERE mc.act_id IN (SELECT act_id FROM movie_cast GROUP BY act_id HAVING COUNT(*) >= 2);</pre>	<table><tr><th></th><th>mov_title character varying (100) 🔒</th></tr><tr><td>1</td><td>MANASU</td></tr><tr><td>2</td><td>AAKASHAM</td></tr><tr><td>3</td><td>KALIYONA</td></tr></table>		mov_title character varying (100) 🔒	1	MANASU	2	AAKASHAM	3	KALIYONA
	mov_title character varying (100) 🔒									
1	MANASU									
2	AAKASHAM									
3	KALIYONA									
28	SQL <pre>SELECT DISTINCT a.act_nam FROM actor a JOIN movie_cast mc1 ON a.act_id = mc1.act_id JOIN movies m1 ON mc1.mov_id = m1.mov_id JOIN movie_cast mc2 ON a.act_id = mc2.act_id JOIN movies m2</pre>	<table><tr><th>act_nam character varying (50) 🔒</th></tr></table>	act_nam character varying (50) 🔒							
act_nam character varying (50) 🔒										

	<pre>ON mc2.mov_id = m2.mov_id WHERE m1.mov_year < 2000 AND m2.mov_year > 2015;</pre>																									
29	SQL <pre>SELECT m.mov_title, COUNT(r.rev_stars) AS number_of_ratings, MAX(r.rev_stars) AS highest_stars FROM movies m JOIN rating r ON m.mov_id = r.mov_id GROUP BY m.mov_title ORDER BY m.mov_title;</pre>	<table><tr><th></th><th>mov_title character varying (100)</th><th>number_of_ratings bigint</th><th>highest_stars integer</th></tr><tr><td>1</td><td>AAKASHAM</td><td>1</td><td>2</td></tr><tr><td>2</td><td>HOME</td><td>1</td><td>3</td></tr><tr><td>3</td><td>KALIYONA</td><td>1</td><td>5</td></tr><tr><td>4</td><td>MANASU</td><td>1</td><td>4</td></tr><tr><td>5</td><td>WAR HORSE</td><td>1</td><td>4</td></tr></table>		mov_title character varying (100)	number_of_ratings bigint	highest_stars integer	1	AAKASHAM	1	2	2	HOME	1	3	3	KALIYONA	1	5	4	MANASU	1	4	5	WAR HORSE	1	4
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Conclusion:

In this experiment, I successfully implemented PostgreSQL commands for database creation and manipulation. DDL commands were used to create tables and define primary and foreign-key constraints, while DML and DQL commands were used for inserting, updating, deleting, filtering, sorting and retrieving records. JOIN, aggregation, grouping and subquery operations were also performed on the movie database. Thus, the experiment provided practical understanding of PostgreSQL commands used for database management and data analysis.

Signature of faculty in-charge with date